

RNA Atlas Explorer — Filters & Metrics

What each selection criterion means, an example it keeps vs one it drops, and how much of the 7,757-fold curated set each removes.

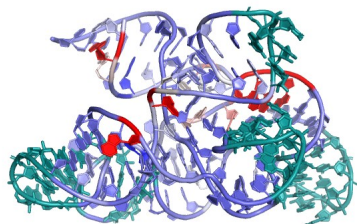
Live: ddc01lh56i5th.cloudfront.net

Length ≥ 40 nt

Short constructs are mostly trivial hairpins; we keep design regions ≥ 40 nt.

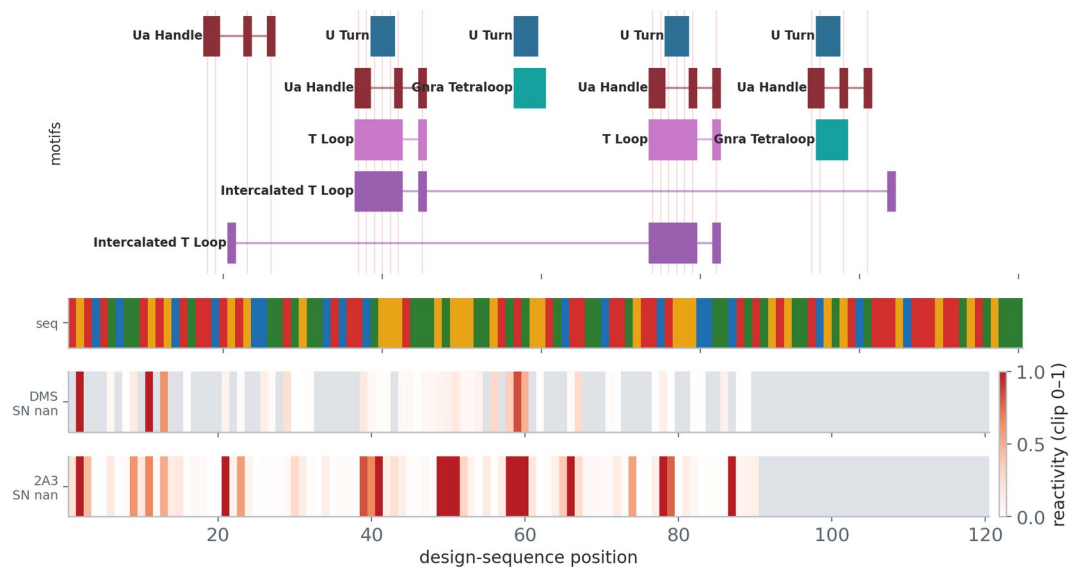
Drops 6,132 of 7,757 folds (79.1%) · keeps 1,625

KEEP — strong example

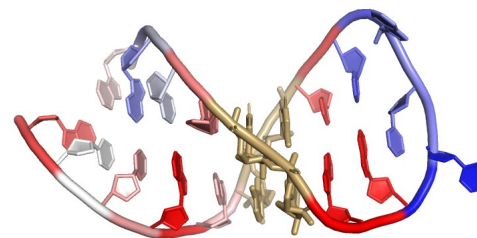


value: 120

0771575-ribonanza2f
natural RNA — RNACentral_130_dedup
120 nt · pLDDT 88
novelty TM1 0.421
nearest 2IHX_B — Solution Structure of the Rous Sarcoma
SHAPE agr 0.3973
compact 0.7167 · paired 0.6167 · 2 tert

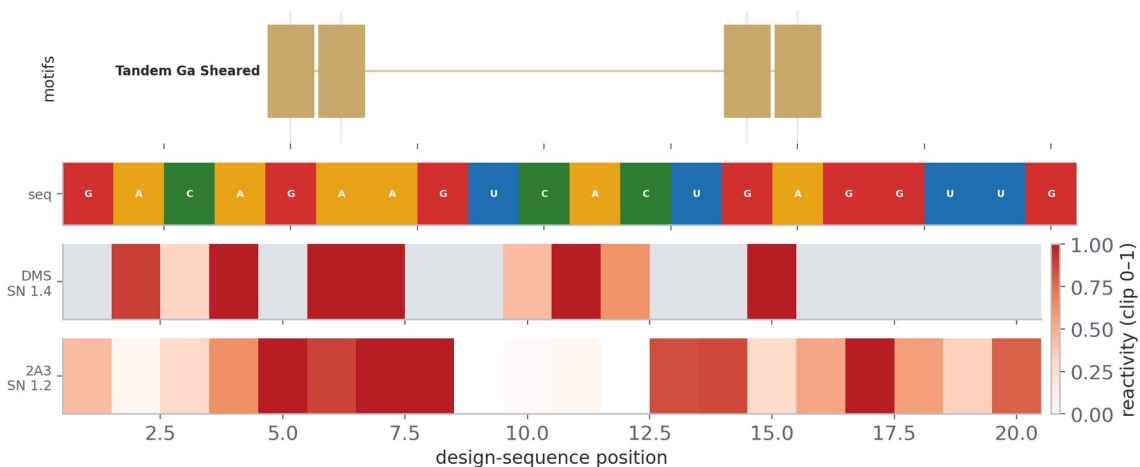


DROP — removed by this filter



value: 20

5355308-ribonanza2e
gRNAde design — target PDB 8FNF_1_m
20 nt · pLDDT 97
novelty TM1 0.437
nearest 6WLH_B — RNA complex of WT1 zinc finger transcr
SHAPE agr -0.1144
compact 0.4 · paired 0.5 · 1 tert

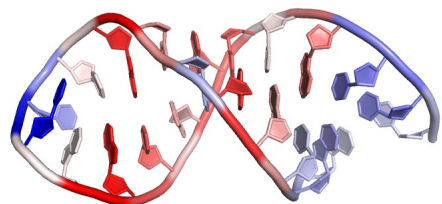


pLDDT ≥ 90

Model confidence. The pool bar is 85; tightening to 90 keeps the most confident folds.

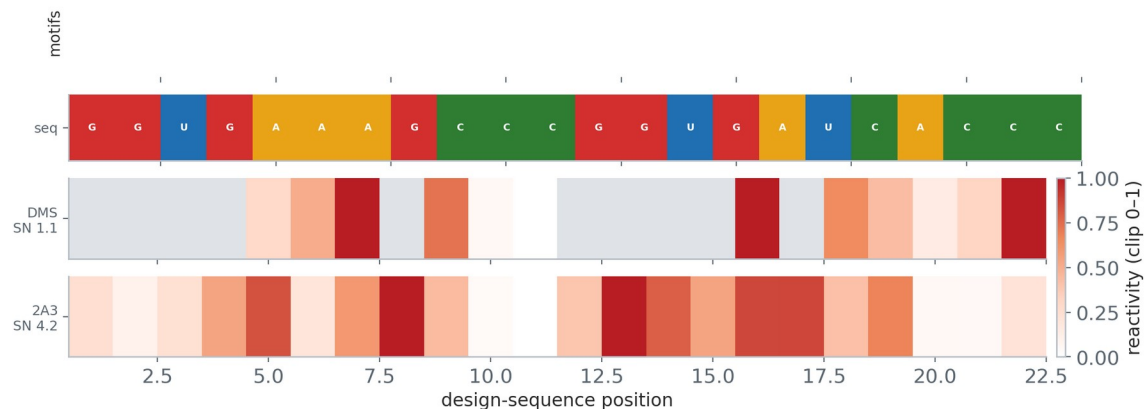
Drops 5,915 of 7,757 folds (76.3%) · keeps 1,842

KEEP — strong example

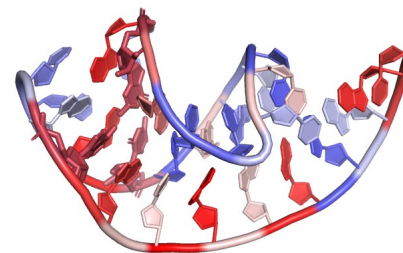


value: 97.763

5932026-ribonanza2a
gRNAde design — target PDB 1KD3_1_A-B
22 nt · pLDDT 98
novelty TM1 0.450
nearest 2L3C_B — Solution structure of ADAR2
dsRBM1 bou
SHAPE agr -0.3351
compact 0.1364 · paired 0.6364 · 0 tert

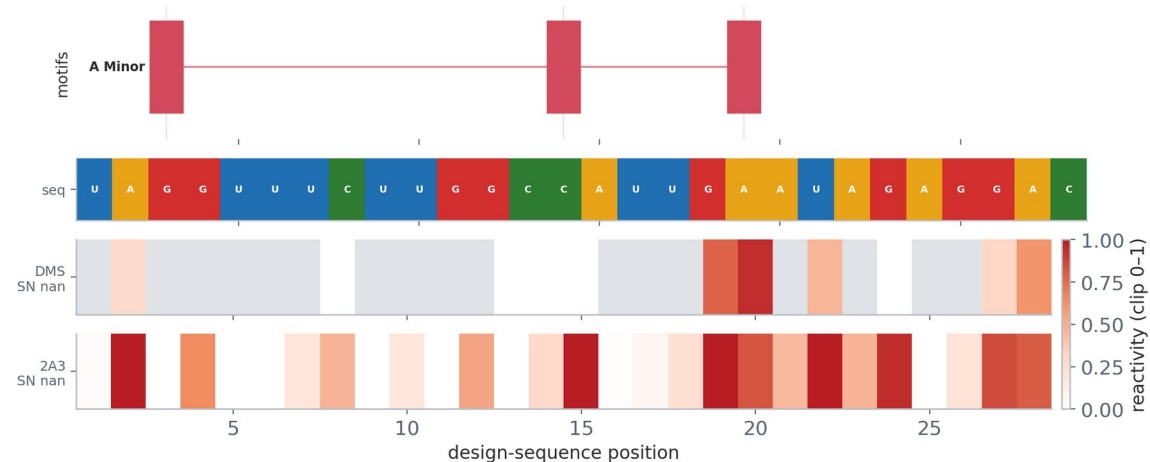


DROP — removed by this filter



value: 85.0

1742929-ribonanza2h
natural RNA · RNACentral100
28 nt · pLDDT 85
novelty TM1 0.373
nearest 1M82_A — SOLUTION STRUCTURE OF
THE COMPLEMENTAR
SHAPE agr 0.2331
compact 0.25 · paired 0.5714 · 1 tert

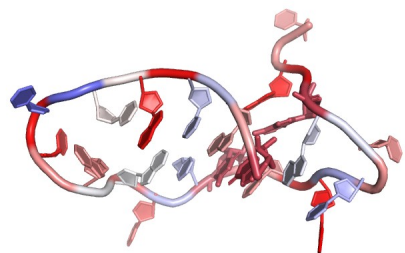


Novelty (best_tm1 ≤ 0.40)

TM1 to the closest v341 PDB-RNA chain; lower = more novel. ≤ 0.40 = clearly unlike known folds.

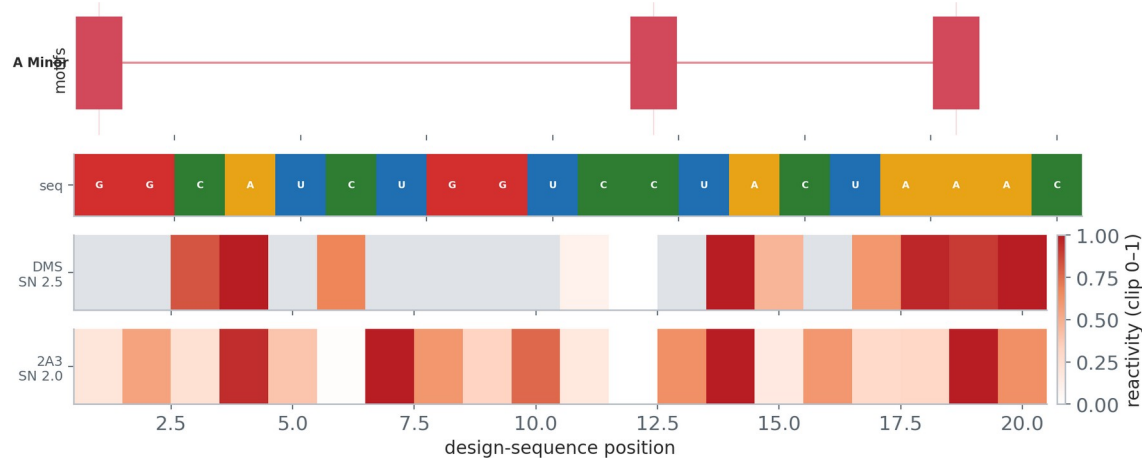
Drops 4,312 of 7,757 folds (55.6%) · keeps 3,445

KEEP — strong example

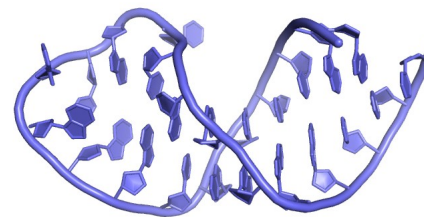


value: 0.0

7214726-ribonanza2d
gRNA design — target PDB 4PQU_1_T
20 nt · pLDDT 92
novelty TM1 0.000
nearest
SHAPE agr 0.5282
compact 0.3 · paired 0.3 · 1 tert

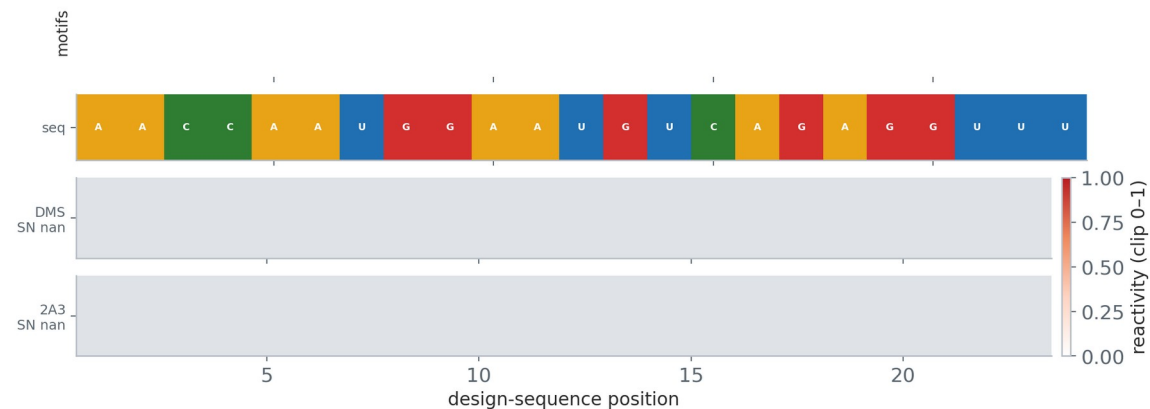


DROP — removed by this filter



value: 0.45

3270371-ribonanza2f
natural RNA · RNACentral_130_dedup
23 nt · pLDDT 96
novelty TM1 0.450
nearest 1RFR_A — NMR structure of the 30mer
stemloop-D
SHAPE agr —
compact 0.3913 · paired 0.5217 · 0 tert

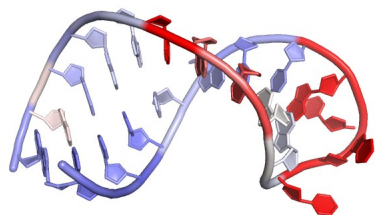


SHAPE agreement ≥ 0.2

Correlation of 2A3 reactivity with unpaired positions; + = mapping supports the predicted fold.

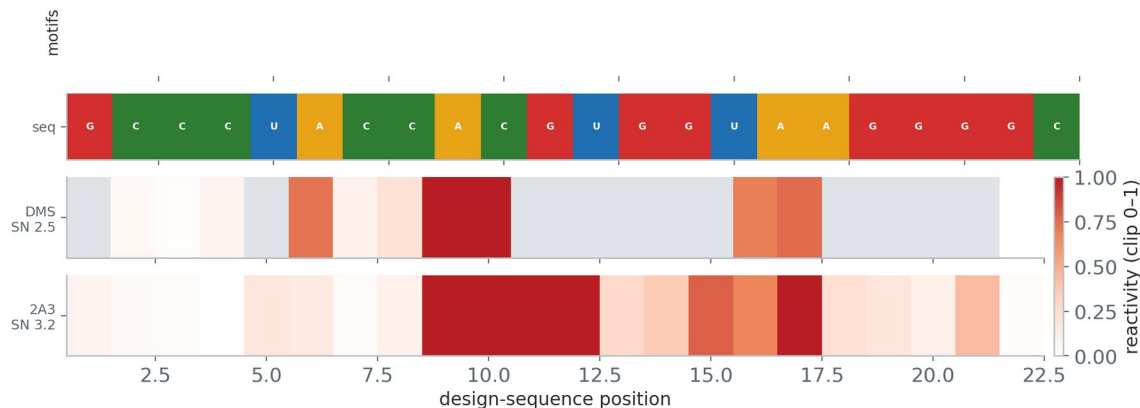
Drops 3,654 of 7,757 folds (47.1%) · keeps 4,103

KEEP — strong example

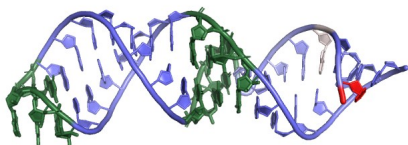


value: 0.877

6900360-ribonanza2e
gRNAde design — target PDB 3CGQ_1_B-A
22 nt · pLDDT 92
novelty TM1 0.433
nearest 2IXY_A — Solution structure of the apical stem-
SHAPE agr 0.8773
compact 0.0455 · paired 0.7273 · 0 tert

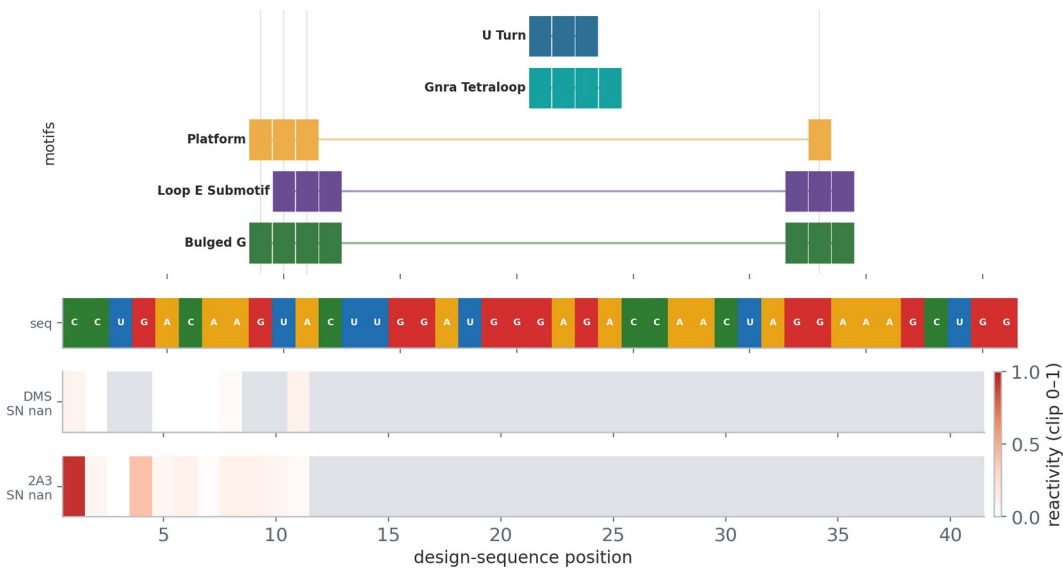


DROP — removed by this filter



value: -0.591

1037281-ribonanza2f
natural RNA · RNACentral_130_dedup
41 nt · pLDDT 92
novelty TM1 0.441
nearest 1MNX_A — The Solution Structure of the Loop E R
SHAPE agr -0.5907
compact 0.439 · paired 0.4878 · 1 tert

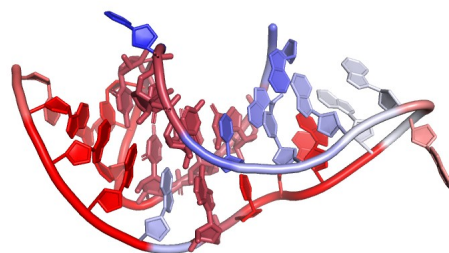


Compactness ≥ 0.4

C1' contact ratio (globularity proxy); higher = more compact, extended chains score near 0.

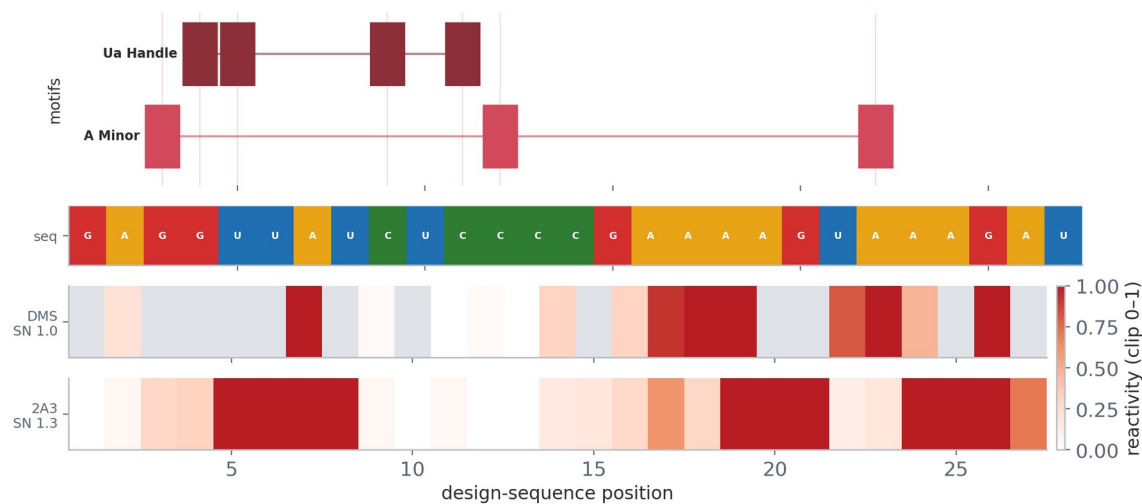
Drops 4,308 of 7,757 folds (55.5%) · keeps 3,449

KEEP — strong example

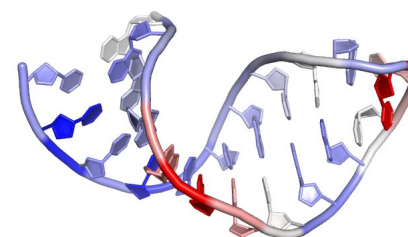


value: 0.852

6701814-ribonanza2c
gRNAde design — target PDB 3K1V_1_A
27 nt · pLDDT 91
novelty TM1 0.384
nearest 2AP0_A — Solution Structure of the
C27A ScYLV P
SHAPE agr 0.1612
compact 0.8519 · paired 0.4444 · 2 tert

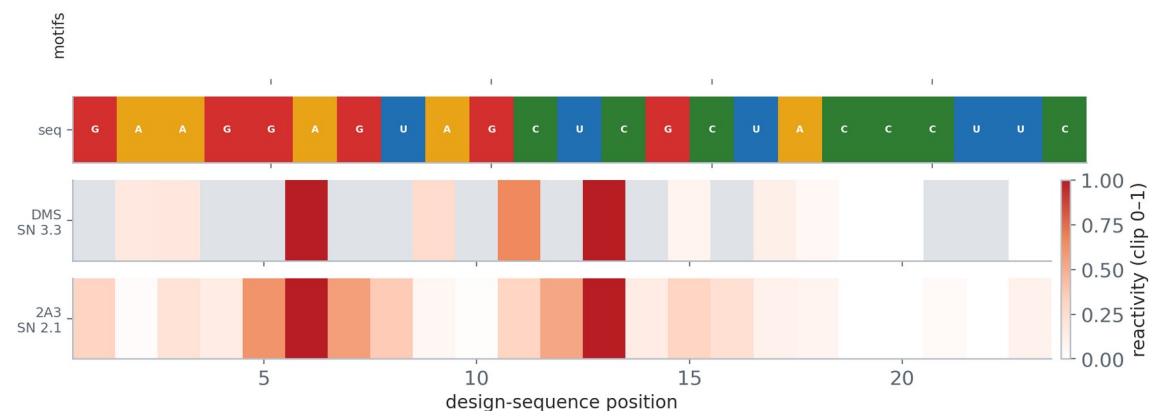


DROP — removed by this filter



value: 0.0

5963544-ribonanza2d
gRNAde design — target PDB 2PJP_1_B
23 nt · pLDDT 97
novelty TM1 0.436
nearest 8QO3_A — Conserved Structures and
Dynamics in 5
SHAPE agr 0.6095
compact 0.0 · paired 0.7826 · 0 tert

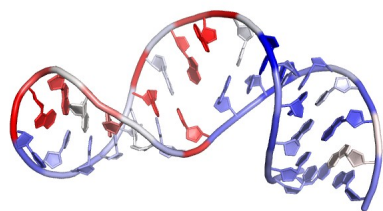


Base-paired ≥ 0.5

Fraction of paired positions in the predicted secondary structure; higher = more structured.

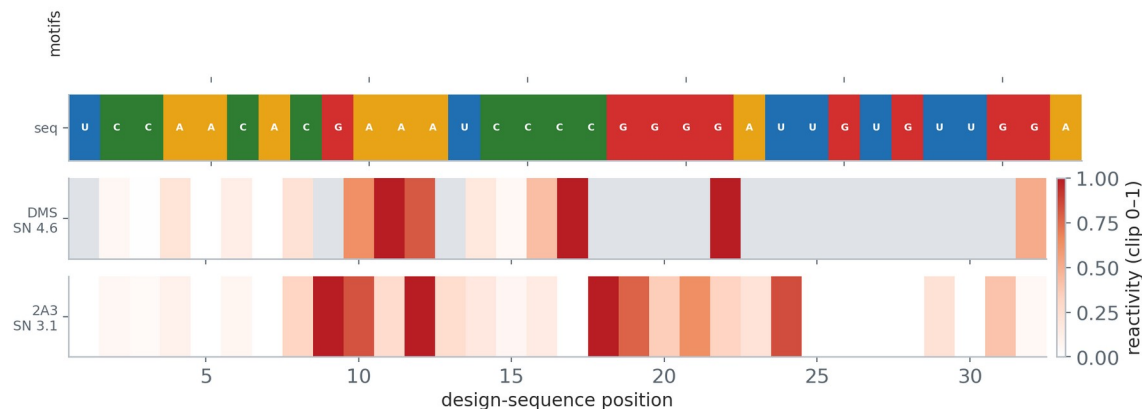
Drops 2,754 of 7,757 folds (35.5%) · keeps 5,003

KEEP — strong example

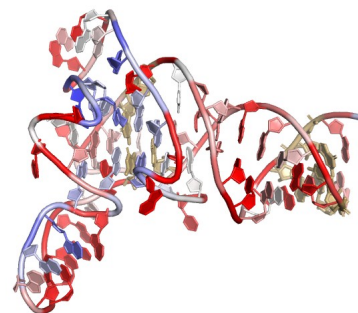


value: 0.875

4427832-ribonanza2d
gRNAde design — target PDB 7KI3_1_F-E
32 nt · pLDDT 87
novelty TM1 0.428
nearest 2L3J_B — The solution structure of the ADAR2 ds
SHAPE agr 0.5364
compact 0.3125 · paired 0.875 · 0 tert

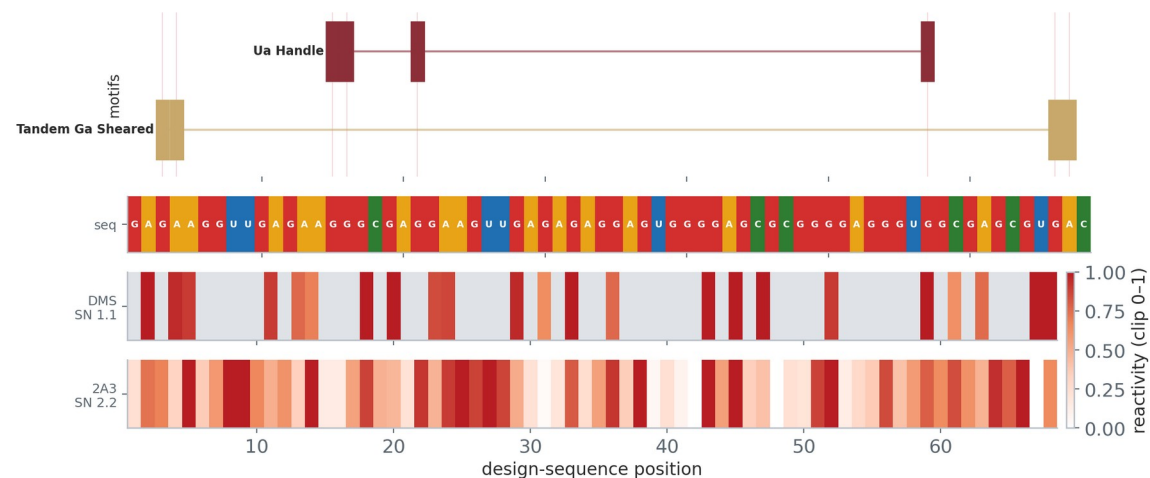


DROP — removed by this filter



value: 0.118

3287527-ribonanza2b
gRNAde design — target PDB 7UWS_1_H
68 nt · pLDDT 85
novelty TM1 0.432
nearest 6VVJ_A — Cap1G-TPUA
SHAPE agr -0.116
compact 0.3824 · paired 0.1176 · 2 tert

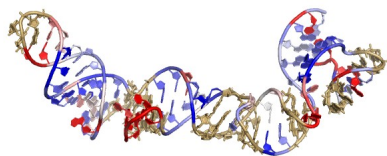


≥ 1 tertiary motif

Carries a Rosetta tertiary motif (A-minor, TL-receptor, T-loop, ...) — a real 3D interaction.

Drops 4,251 of 7,757 folds (54.8%) · keeps 3,506

KEEP — strong example



value: 6

2570713-ribonanza2d

UW design AK_002872

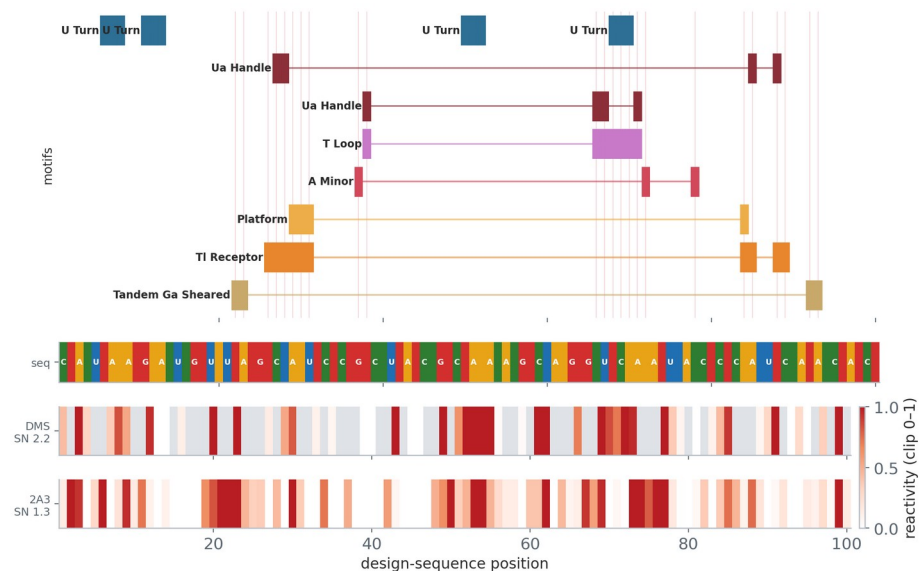
100 nt · pLDDT 85

novelty TM1 0.425

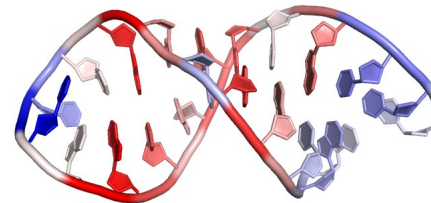
nearest 6VJ_A — Cap1G-TPUA

SHAPE agr 0.473

compact 0.57 · paired 0.52 · 6 tert



DROP — removed by this filter



value: 0

5932026-ribonanza2a

gRNAde design — target PDB 1KD3_1_A-B

22 nt · pLDDT 98

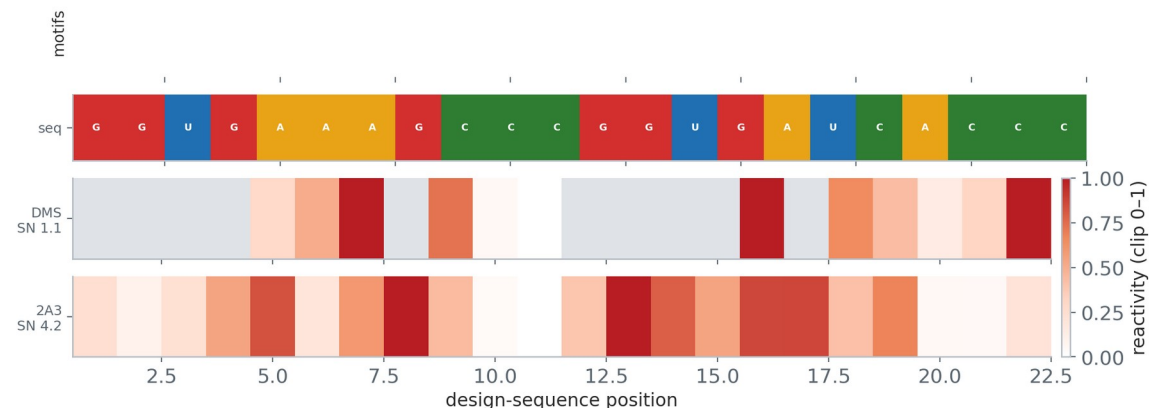
novelty TM1 0.450

nearest 2L3C_B — Solution structure of ADAR2

dsRBM1 bou

SHAPE agr -0.3351

compact 0.1364 · paired 0.6364 · 0 tert



Filter combinations — what survives

All 64 on/off combinations of 6 core filters · sorted by # kept

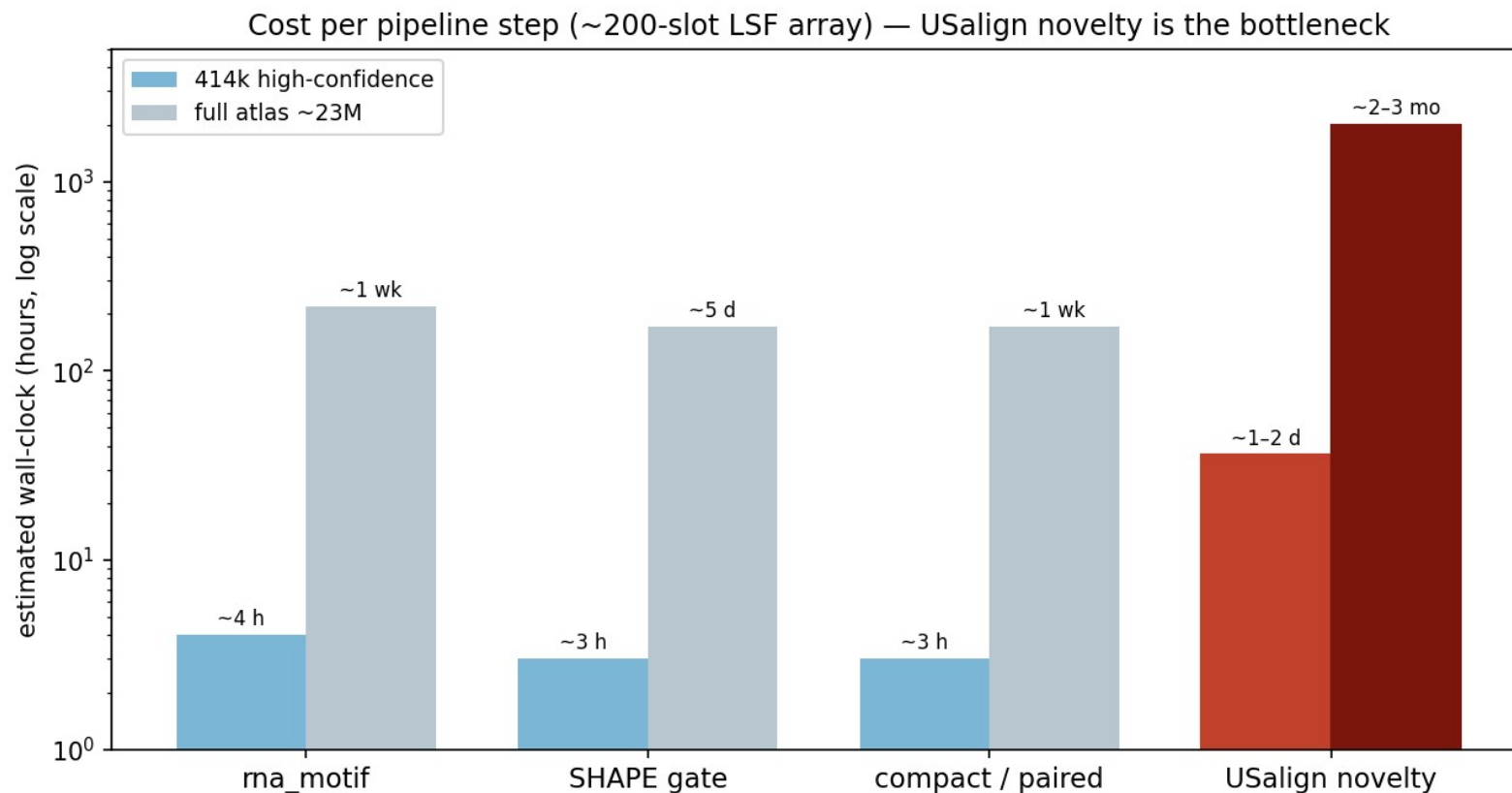
len	pLDDT	novel	SHAPE	compact	motif	% left	# left
.	.	.	.	.	.	100.0	7,757
✓	.	.	✓	.	.	58.3	4,522
.	.	.	.	✓	✓	45.2	3,506
.	.	.	.	✓	.	44.5	3,449
✓	.	✓	.	.	.	44.4	3,445
.	.	.	✓	✓	✓	33.6	2,604
.	.	.	✓	✓	.	30.8	2,389
.	.	.	.	✓	✓	28.5	2,208
.	.	✓	✓	.	.	24.7	1,919
✓	✓	.	.	.	.	23.7	1,842
.	.	✓	✓	✓	✓	23.1	1,789
.	.	✓	.	.	✓	22.0	1,705
✓	.	.	.	.	.	20.9	1,625
✓	.	.	✓	.	.	18.4	1,431
.	.	✓	.	✓	.	18.0	1,398
✓	.	.	.	✓	.	16.5	1,278
.	.	✓	✓	.	✓	15.6	1,213
✓	.	.	.	✓	✓	15.6	1,208
✓	.	.	✓	✓	.	15.0	1,161
✓	.	.	✓	.	✓	14.2	1,100
.	✓	.	✓	.	.	14.1	1,093
✓	.	.	.	✓	✓	13.7	1,066
✓	.	.	✓	✓	✓	12.7	984
.	.	✓	✓	✓	.	12.0	932
.	✓	✓	✓	.	.	11.8	916
.	✓	✓	.	.	✓	11.5	889
.	✓	✓	.	.	.	10.8	836
.	✓	✓	.	✓	.	10.6	824
.	✓	✓	✓	✓	✓	9.2	712
.	✓	✓	✓	.	✓	8.1	626
.	✓	✓	.	.	✓	6.7	523
.	✓	✓	✓	.	.	6.7	523
.	✓	✓	✓	✓	.	6.7	516
.	✓	✓	.	✓	✓	6.4	497
.	✓	✓	.	.	.	5.1	394
.	✓	✓	✓	.	✓	5.0	386
✓	.	✓	.	.	.	4.8	373
.	✓	.	✓	✓	✓	4.7	367
✓	.	✓	✓	.	.	4.2	322
.	✓	✓	.	✓	✓	3.6	279
✓	.	✓	.	✓	.	3.4	267
✓	.	✓	.	.	✓	3.4	263
.	✓	✓	✓	✓	.	3.2	250
✓	.	✓	✓	.	.	3.1	241
✓	.	✓	✓	.	✓	3.1	239
✓	.	✓	.	✓	✓	2.8	221
✓	.	✓	✓	✓	✓	2.6	205
.	✓	✓	✓	✓	✓	2.6	204
✓	✓	.	.	.	.	1.2	90
✓	✓	.	✓	.	.	1.1	82
✓	✓	.	.	✓	.	0.9	71
✓	✓	.	✓	.	.	0.9	67
✓	✓	.	.	.	✓	0.8	65
✓	✓	.	✓	✓	✓	0.8	61
✓	✓	.	.	✓	✓	0.8	59
✓	✓	✓	✓	✓	✓	0.7	56
✓	✓	✓	.	.	.	0.2	16
✓	✓	✓	.	✓	.	0.2	14
✓	✓	✓	✓	.	.	0.2	14
✓	✓	✓	✓	✓	.	0.2	13
✓	✓	✓	.	.	✓	0.2	12
✓	✓	✓	.	✓	✓	0.2	12
✓	✓	✓	✓	.	✓	0.1	11
✓	✓	✓	✓	✓	✓	0.1	11

How to read

- Each row = one on/off combination of the 6 core filters (✓ = applied).
- All 64 combinations; top row = no filters (all 7,757).
- “% left” / “# left” = folds passing every applied filter (AND).
- Sorted by # kept — tightest combinations at the bottom.
- Filters overlap, so stacking them removes less than the sum of each alone.
- Full table (incl. all 8 filters): combinations.csv.

Scaling to the full predicted atlas — cost & bottleneck

Everything here is computed for the 7,757 curated folds. Running the same pipeline atlas-wide is gated by one step.



Bottleneck

USalign novelty: ~0.5–3 min/fold. Every other step is seconds or sub-second.

Feasible

414k high-confidence set: ~1–2 days on a ~200-slot LSF array.

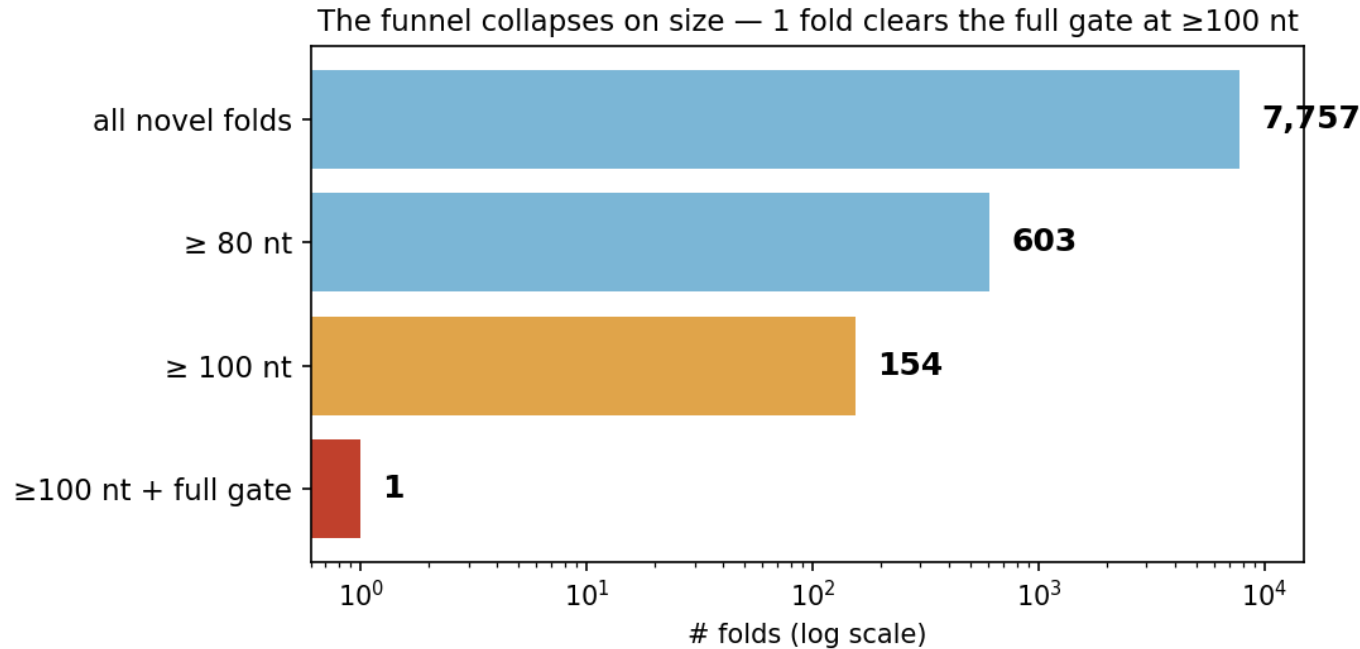
Impractical

Full ~23M atlas: novelty alone is ~2–3 months on a 200-slot array — and most are low-confidence. Even the cheap steps become week-scale.

Recommended

Run the full pipeline over the 414k strict index; skip novelty on the low-confidence rest (or throw far more slots at it).

Target selection collapses on size



The hard reality

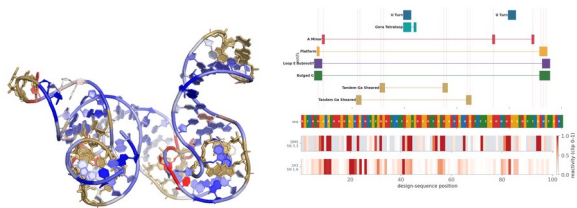
- Un-assisted single-particle cryo-EM of RNA wants $\sim 150+$ nt (~ 50 kDa).
 - Ribonanza-2 design regions cap ~ 130 nt — so every candidate is small for cryo-EM.
 - Of 7,757 novel folds, only 154 are ≥ 100 nt and just 1 clears the full quality gate.
- **The decision that sets the list:**
- Accept Fab/scaffold-assisted cryo-EM? If yes → hundreds of small but striking folds open up.
 - If no → we're limited to the ~ 12 largest, fully de-risked folds.

Proposed list — two routes (rank, then curate together)

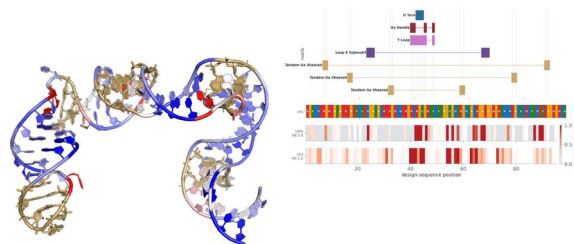
Route A • Largest, fully de-risked (12)

len ≥ 80 · pLDDT ≥ 88 · SHAPE-supported · compact ≥ 0.5 · novel (TM1 ≤ 0.40) · ≥ 1 tertiary

Best shot at a usable map with least scaffolding; mostly RNAMake assemblies ~90–100 nt.



4146783-ribonanza2e • 100 nt
RNAMake assembly (from 4WT8, 6TH6)
tm 0.40 · agr 0.5321 · comp 0.62 · 3t



7979215-ribonanza2d • 95 nt
RNAMake assembly (from 1NJP, 5J7L)
tm 0.40 · agr 0.5029 · comp 0.6632 · 3t

More Route A candidates:

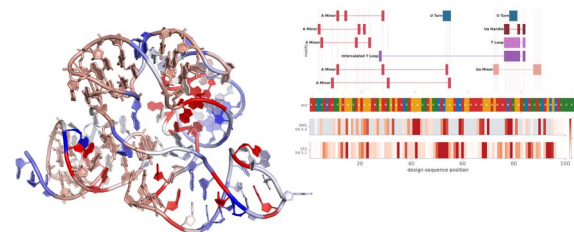
0900292-ribonanza2d 95 nt · RNAMake assembly (from 5MMM, 6ZVI) · tm 0.39 · agr 0.4749 · comp 0.5789 · 2t

5868329-ribonanza2d 94 nt · RNAMake assembly (from 5E81, 7ST2) · tm 0.38 · agr 0.4766 · comp 0.5319 · 1t

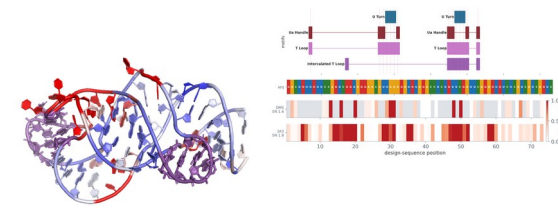
Route B • Most striking architecture (103)

novel · SHAPE-supported · pseudoknot · rare tertiary motif

Smaller → cryo-EM needs a chaperone/scaffold, but high scientific interest.



3066731-ribonanza2d • 100 nt
UW design AK_002812_2098945_003
tm 0.38 · agr 0.3637 · comp 0.59 · 4t · PK



7803325-ribonanza2c • 73 nt
RNAMake assembly (from 5ZEB, 6ZJ3)
tm 0.37 · agr 0.5992 · comp 0.6027 · 2t · PK

More Route B candidates:

4286710-ribonanza2c 24 nt · gRNAd design — target PDB 6W62_1_B · tm 0.40 · agr 0.4968 · comp 0.4167 · 3t · PK

5993836-ribonanza2e 72 nt · RNAMake assembly (from 6JXM, 6YW5) · tm 0.38 · agr 0.4076 · comp 0.6667 · 2t · PK

3D colored by 2A3 reactivity (blue protected→red reactive) + motif sticks; tracks = sequence · DMS · 2A3 · motifs. Mostly synthetic designs — large+SHAPE+novel natural pool is near-empty. Pull ~5–8/route → ~10–15 portfolio.

What de-risks a cryo-EM slot — and the axis we're missing

cryo-EM is expensive — each metric buys a piece of confidence before we commit:

- **Model says it folds** — pLDDT + gPDE.
- **Experiment says it folds (orthogonal!)** — SHAPE agreement — guards against a confidently-wrong model.
- **It's a compact single particle** — compactness (C1' contact ratio) + base-paired fraction.
- **It's worth solving** — novelty vs the PDB + the nearest-entry title as a sanity check.
- **It's interesting** — tertiary motifs, pseudoknots.

The axis we don't yet measure: rigidity / single dominant conformation

A floppy, multi-conformer RNA gives no usable map regardless of how it scores above. We have one prediction per fold, so we can't see conformational spread.

Proposal: before committing, re-predict each shortlisted fold with N seeds and keep the low inter-seed-RMSD (single rigid fold) ones — cheap for a ~10–30 shortlist.

Open questions for the discussion

1. What size can the cryo-EM team realistically attempt — and are they open to Fab / scaffold assistance for $< \sim 150$ nt?
2. RNA-only targets, or RNA-protein complexes?
3. Natural (F-H, more reliably folded) vs synthetic (A-E, more novel) — and what mix?
4. Hard-require experimental SHAPE support, or allow model-only candidates?
5. How many targets are we scoping — ~ 5 hero targets, or a ~ 20 – 30 shortlist?

Proposal: bring the explorer to the meeting, tune thresholds live, and walk the top of the ranked list together.